

CEBU INSTITUTE OF TECHNOLOGY – UNIVERSITY

COLLEGE OF COMPUTER STUDIES

Software Requirements Specification (SRS)

for

Smart Document Evaluator (Smart Docs Validator) with AI Integration

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Document version: 3.1 (deliverable revision)

Date: May 14, 2026

Note on templates: The department Word templates (`SRS_TEMPLATE.doc` , etc.) could not be imported as text (binary `.doc` / `.docx`). This Markdown file follows the **same conventional SRS outline** (IEEE 830–style sections) and is the **authoritative v3.1 requirements baseline** for the repository as deployed.

Change History

Version	Date	Description
1.0–2.x	2025–2026	Prior iterations; see <code>docs/SRS-Smart-Document-Evaluator-v2.md</code> .
3.0	May 13, 2026	Baseline aligned with Vite + React + Supabase + Vercel + Gemini (see v2 file header).
3.1	May 14, 2026	Amendments: Teacher Course Tasks (published assignments, optional handouts, due dates, delete/bulk-delete); student Tasks / Submit Work deep links (<code>openTask</code>); deadline reminders (in-app + optional browser notifications); General Submission bucket hidden from student task/workspace lists (still used for quick submit); STD document type on course tasks; nav label updates. Supersedes contradictory “teachers do not create assignments” wording from v3.0 where it conflicts with this revision.

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1. Introduction

1.1 Purpose

This SRS specifies functional and non-functional requirements for **Smart Document Evaluator** (UI brand: **Smart Docs Validator**), a web application for **IT332 / CS342** at CIT–University. Students submit coursework; teachers review AI-assisted evaluations and publish grades; optional **Google Gemini** provides rubric-aligned feedback.

Audience: developers, advisers, QA, deployers, and academic reviewers.

1.2 Scope

In scope

- Google OAuth (Supabase Auth) with **class-list access gate** and role mapping (student , teacher , admin).
- Student **Submit Work**, **Submission Status**, **Tasks**, **Boards**, **Calendar**, **Drive**, **Sheets**, **Analytics**, **Team 14**, **Settings**; onboarding tour (**Eva**); optional **Rate us** (Google Forms).
- Teacher **Dashboard**, **Grades** (review queue), **Student Submissions**, **Documents**, **Analytics**, **Class List**, **Instructions/Inbox**, **Settings**, **Team 14**, **Reports**.
- **Course Tasks**: teachers create published tasks (title, instructions, document type **SRS | SDD | SPMP | STD | Other**), optional **due date/time**, optional **handout** file (Supabase Storage); students see active tasks on **Tasks** and may open **Submit Work** with a task pre-selected.
- **AI evaluation** (Gemini or heuristic fallback), **two-lane** AI vs teacher grading, **redo** workflow, **CSV** export, **invitation email** pipeline (Resend + CLI fallbacks).
- Deployment on **Vercel** + data on **Supabase** (Postgres + Storage + RLS).

Out of scope

- Official LMS replacement; native mobile app stores; legal certification of AI output as sole evidence of integrity.

1.3 Definitions, acronyms, and abbreviations

Term	Definition
SPA	Single-page application (React + Vite + TypeScript).
RLS	Row-Level Security on Postgres.
Course task	Teacher-authored assignments row shown to students (vs. system “General Submission” bucket).
General Submission	Auto-provisioned assignment for quick uploads without selecting a named task; hidden from student Tasks/workspace lists but still used server-side.
Gemini proxy	Same-origin POST /api/gemini-evaluate using server GEMINI_API_KEY.

1.4 References

- IEEE Std 830-1998 — SRS recommended practice.
- docs/SDD-Smart-Documents-Evaluator-v3.md — design description.
- docs/SRS-Smart-Documents-Evaluator-v2.md — detailed FR baseline (see §3.1).
- README.md , docs/supabase-setup-all-in-one.sql , Google Gemini / Supabase / Vercel / Resend documentation.
- Republic Act No. 10173 — Data Privacy Act of 2012 (Philippines).

1.5 Overview

Section 2 describes users, constraints, and dependencies. Section 3 states **baseline** requirements (v2) and **v3.1 amendments**. Section 4 maps amendments to design/tests.

2. Overall description

2.1 Product perspective

Standalone HTTPS web app: browser → Vercel (SPA + api/*) → Supabase (Auth, Postgres, Storage) → optional Google Gemini and Resend. See architecture diagram in README.md and docs/SDD-Smart-Document-Evaluator-v3.md .

2.2 User characteristics

- **Student** — submits files, tracks status, uses Tasks/Boards; may receive deadline reminders.
- **Teacher** — manages class list, runs AI grading, publishes scores, manages **Course Tasks** (create/close/delete).
- **Admin** — env-driven elevated access; same tooling as teacher where applicable.

2.3 Constraints

- Google sign-in only; roster allow-list for students; VITE_TEACHER_EMAILS / VITE_ADMIN_EMAILS for staff.
- File size and type limits per README.md (e.g. ≤ 25 MB).
- Production Gemini key must not ship in client bundle (use /api/gemini-evaluate).

2.4 Assumptions and dependencies

- Stable internet; valid Supabase, Vercel, and (optional) Resend/Gemini credentials.
- Students use Google accounts listed in the class-list source used by the app gate.

3. Specific requirements

3.1 Baseline functional requirements (SRS v2)

The **numbered functional requirements FR-01 ...** and interface requirements **SI-1 ...** in:

docs/SRS-Smart-Document-Evaluator-v2.md

remain valid **except** where they explicitly state that teachers **do not** create or manage assignments. That sentence is **superseded** by §3.2 below for **Course Tasks** only. All other FR in v2 continue to apply (authentication, submissions, AI lane, teacher lane, invitation email, analytics, etc.).

3.2 Amendments and new requirements (v3.1)

ID	Requirement
AM-01	The system shall allow a teacher to create a course task with: required title; optional description; document type in { srs, sdd, SPMP, STD, other }; optional due date/time; optional handout file.
AM-02	On successful create, the system shall persist the task in assignments (or assignment if singular table) with teacher_id = current user and status = active by default.
AM-03	If a handout file is provided, the system shall upload it to the configured Storage bucket under a teacher-scoped path and shall persist handout_url and handout_file_name when those columns exist (otherwise show a documented SQL migration path).
AM-04	The system shall allow the teacher to close or re-open a course task (status closed/active).
AM-05	The system shall allow the teacher to delete a single course task and to delete selected tasks (bulk), with confirmation, scoped to rows owned by that teacher.
AM-06	Students shall see all active course tasks assigned to the cohort in Tasks (grouped by due-date urgency) and in the shared student workspace data feed, excluding rows whose title matches the configured General Submission bucket title (quick-submit bucket remains available via Submit Work flows).

AM-07	From Tasks , the system shall provide a path to Submit Work with query <code>openTask=<assignmentId></code> so the turn-in modal opens for that task.
AM-08	The system shall surface in-app deadline reminders for active tasks with due dates within a defined near window (e.g. 72 hours) or overdue, with dismiss persistence in <code>sessionStorage</code> where implemented.
AM-09	When the browser grants permission, the system may emit browser notifications for imminent deadlines subject to per-assignment daily deduplication.
AM-10	Database check constraint on <code>assignments.document_type</code> shall include STD alongside SRS, SDD, SPMP, Other (migration script provided in repo for existing databases).
AM-11	The Submit Work UI shall allow clearing a selected handout file before publish and shall provide a due-date picker with explicit OK confirmation when implemented as a custom control.

3.3 Non-functional requirements (additions)

ID	Requirement
NFR-AM-01	Course task delete operations shall respect RLS (<code>assignments_delete_own</code> or equivalent).
NFR-AM-02	Hiding the General Submission row from student workspace views shall not break quick submit or submission rows referencing that assignment id.
NFR-AM-03	Analytics completion metrics shall remain consistent when workspace-visible assignments are a subset of all assignment ids (exclude hidden bucket from denominators where applicable).

4. Requirements traceability

Amendment	SDD section	STD section
AM-01-05	§3.8 Course Tasks (Teacher)	TC-CT-01 ... TC-CT-08
AM-06-09	§3.2 Student Submission; reminders component	TC-ST-01 ... TC-ST-06
AM-10	§4 Data Design; SQL docs	TC-DB-01
AM-11	Teacher modal UI	TC-UI-01

5. References

- docs/STD-Smart-Document-Evaluator-v3.md
- docs/SPMP-Smart-Document-Evaluator-v3.md
- docs/SDD-Smart-Document-Evaluator-v3.md
- docs/PRESENTATION-READINESS-AND-DELIVERABLES.md